

COMPUTER NETWORKS LABORATORY

CN LAB – EXPERIMENT 16**Do the Following Using NS2 Simulator**

Practical Record • GitHub Submission

Experiment	16
Lab Topic	Do the following using NS2 Simulator
Environment	NS-2 (Network Simulator 2)
Purpose	Network simulation and packet analysis
Submission	GitHub / LMS

Note: This record is formatted from the NS2 Simulator lab task shown in the provided LMS screenshot.

1. Aim

To perform network simulations using the NS2 simulator and analyze packet transmission and packet loss in a simulated network.

2. Lab Tasks

No.	Task
1	NS2 Simulator – Introduction.
2	Simulate to find the number of packets dropped.
3	Simulate to find the number of packets dropped by TCP/UDP.
4	Simulate to find the number of packets dropped due to the specified network condition.

3. NS2 Simulator – Introduction

NS2 (Network Simulator 2) is a discrete-event network simulation tool used to model and study communication networks. It can simulate protocols, network topologies, traffic, queues, packet transmission and packet loss.

4. Basic Simulation Workflow

Step	Procedure
1	Create the required network topology using an NS2 Tcl script.
2	Configure links, bandwidth, delay and queue parameters.
3	Create the required traffic sources and transport protocols.
4	Start and stop traffic according to the simulation scenario.
5	Run the Tcl simulation using NS2.
6	Analyze the trace output to determine transmitted, received and dropped packets.

5. Packet Drop Analysis

Packet drops can occur when a network queue becomes full, when packets are discarded by a protocol, or when a simulated network condition causes packet loss. The NS2 trace file can be analyzed to count packet events.

6. TCP / UDP Packet Drop Analysis

TCP and UDP can be compared by creating separate simulation scenarios or traffic flows. The resulting trace data can be examined to determine how many packets were transmitted, received and dropped for each protocol.

7. Expected Observation

Parameter	Observation
Packets Sent	Total packets generated by the traffic source.
Packets Received	Packets successfully delivered to the destination.
Packets Dropped	Packets discarded during the simulation.
Packet Loss	Difference between transmitted and successfully received packets.

8. Result

The NS2 simulator can be used to model network communication and determine packet transmission and packet-drop behavior. TCP and UDP traffic can be simulated and their packet-level results can be analyzed from the generated trace data.

9. Precautions

- Use the correct NS2 Tcl script and simulation parameters.
- Check the trace file before calculating packet statistics.
- Keep the network topology and traffic configuration consistent when comparing TCP and UDP.
- Verify the simulation duration and traffic start/stop times.